

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	M. VAISHNAV	Reg. No.:	192524251
Section / Batch:		Date:	20/07/2026

Code of Conduct

I M. VAISHNAV C192524251 (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

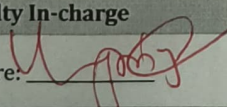
- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:		100 Marks	

38	192524251	VAISHNAV M	- Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. - A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	- Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. - A queue implemented using an array report "Queue Full" even though some positions are empty. - Analyze why this occurs. - Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	- Consider the singly linked list: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50$. Delete the node containing 30. Explain how the links change after deletion. - Given two sorted linked lists: L1: $1 \rightarrow 3 \rightarrow 5 \rightarrow 7$ L2: $2 \rightarrow 4 \rightarrow 6 \rightarrow 8$

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

(CSA-0305) CO2-AT1 (Analytical Problem)

Q1. Comparison Between Recursion and Explicit Stack

Here, Two algorithms can solve the same problem using either recursion or an explicit stack.

Their comparison is as follows:-

Aspect	Recursion (Algorithm A)	Explicit Stack (H.B)
Memory usage	uses the system call stack automatically. Each function call creates a stack frame.	uses a user-defined stack data structure in memory.
Execution flow	Simpler and easier to understand. Function calls and returns are managed automatically.	More control over execution. programmer manually performs push and pop operations.
Risk of stack overflow	High for deep recursion because the system call stack has limited size.	Lower risk because stack size can be managed dynamically.
Implementation complexity	shorter and more readable code.	slightly longer and more complex code.
Performance	Extra overhead due to repeated function calls.	usually faster because it avoids function call overhead.

Conclusion :-

Recursion is easier to write and understand, while an explicit stack provides better control over memory.

Q2. stack overflow in a stack of capacity 5
given operations;

* push(A) * push(B) * push(C) * push(D)
* push(E) * push(F).

step - by - step Execution :-

operation	stack contents
push(A)	A
push(B)	A, B
push(C)	A, B, C
push(D)	A, B, C, D
push(E)	A, B, C, D, E
push(F)	overflow Error

Explanation :-

The stack capacity is 5 elements.
after inserting A, B, C, D and E, the stack becomes full.

TOP :-

E

D

C

B

A

when push(F) is attempted, there is no available space in the stack. Therefore, a stack overflow error occurs.

Error Identified :-

Stack overflow - occurs when an insertion (push) operation is attempted on a full stack.

Two possible solutions :-

Solution 1 :-

Increase stack capacity

Allocate more ~~memory~~ or create a larger stack so ~~so~~ additional elements can be stored.

Solution 2 :-

Remove elements Before Pushing
use one or more pop() operations to
remove existing elements and create space before
inserting F.

Example :-

~~pop()~~ → Removed E
~~push(F)~~ push(F)

Resulting stack :-

TOP :-

F
D
C
B
A

Conclusion :-

The operation push(F) caused a stack overflow because the stack has reached its maximum capacity of 5 elements.